

ENERGY SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military Action Extension Methodology

DBA-EN-SF: Energy Sector Parent Document

*Governing Parent for DBA-ES, DBA-GAS, DBA-OIL,
and All Sub-Sector Instruments in the Energy Sector Series*

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For review by engineers, regulators, asset owners, and policy analysts responsible for protecting energy sector critical infrastructure across the electric, natural gas, and petroleum sub-sectors.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the NIPP Energy sector. It is the parent instrument for three sub-sector series: DBA-ES (Electric sector), DBA-GAS (Natural Gas sector), and DBA-OIL (Petroleum sector). Each sub-sector series inherits the methodology and organizing principles established here and applies them to the specific facility types, operational characteristics, and regulatory contexts of its sector.

The Energy sector, as defined by the National Infrastructure Protection Plan (NIPP), encompasses the electric power system, natural gas production and delivery infrastructure, and petroleum production and refining infrastructure. DHS is the Sector-Specific Agency (SSA) for the Energy sector, with DOE, FERC, NERC, TSA, and PHMSA as the primary federal regulatory and operational partners depending on sub-sector and function.

This document applies the DB → DBA → DBA-MA three-layer analytical progression at the Energy sector level: a baseline Design Basis (DB) establishing the postulated conditions the sector's facilities must withstand; a Design Basis Accident (DBA) layer bounding the non-adversarial consequence envelope; and a military action extension (DBA-MA) applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to the sector's governing postulated events. The sector-level instruments establish this progression as a requirement; each sub-sector framework and facility-class instrument develops it in detail.

1.2 What This Document Does Not Do

This document does not supersede or modify any existing sub-sector framework or facility-class instrument. DBA-ES, DBA-GAS, and DBA-OIL are established series with their own governing documents; this parent establishes the common architecture they share and the cross-sector cascade analysis that spans their boundaries.

This document does not rename the sub-sector series. DBA-ES, DBA-GAS, and DBA-OIL retain their existing designations. The DBA-ES series retains its internal naming convention (DBA-ES-HY-FC1, DBA-ES-TX-FC2, etc.) without modification.

This document does not create new facility class designations. FC designations are the authority of their respective sub-sector frameworks. This document carries a series of inventory tables that reflect the current FC register state across all three sub-sectors.

1.3 Inductive Architecture Principle

The Energy sector series is built inductively: parent documents emerge from mature children, not the reverse. DBA-EN-SF v0.3 is drafted at this version because the three sub-sector frameworks have now reached sufficient maturity to reveal the common architecture they share. This

document does not direct the sub-sector frameworks; it reflects and codifies what they have established.

Future revisions of DBA-EN-SF will be triggered by substantive developments in the sub-sector frameworks — new facility classes, new cross-sector cascade methodologies, or changes to the governing regulatory context — not by a top-down revision schedule.

1.4 Sub-Sector Series Inventory

Sub-Sector Series	Sector Framework	Current Version	SSA	Active FC Instruments
DBA-ES (Electric)	DBA-ES-Sector_Framework	v0.10 — DRAFT	DOE / FERC / NERC	DBA-ES-HY-FC1 (v0.5), DBA-ES-TX-FC2 (v0.3), DBA-ES-TX-FC3 (v0.3), DBA-ES-GC-FC4 (v0.5), DBA-ES-DI-FC5 (v0.4), DBA-ES-CntlCo (v0.3), Critical Loads subsector (DBA-ES-CL): DBA-DC (v1.9), DBA-ES-GN-FC6 (v0.2), DBA-ES-GN-FC7 (v0.2), DBA-ES-GN-FC8 (v0.2), DBA-ES-POL (v0.1), DBA-ES-MA-SF (v0.2), DBA-ES-MA-QB (v0.1)
DBA-GAS (Natural Gas)	DBA-GAS-Sector_Framework	v1.7 — DRAFT	DOE / FERC / TSA / PHMSA	DBA-GAS-FC1 (v1.6), DBA-GAS-FC2 (v0.2), DBA-GAS-FC3 (v0.1), DBA-GAS-FC4 (v0.1), DBA-GAS-FC-LNG (promoted; annex DBA-GAS-UC-LNG v1.1), DBA-GAS-POL (v1.3), DBA-GAS-MA-SF (v0.2)
DBA-OIL (Petroleum)	DBA-OIL-Sector_Framework	v1.8 — DRAFT	DOE / TSA / PHMSA / EPA	DBA-OIL-FC1 (v1.3), DBA-OIL-FC2 (v0.1), DBA-OIL-FC3 (v0.1), DBA-OIL-FC4 (v0.3), DBA-OIL-FC5 (promoted), DBA-OIL-MA-SF (v0.1)
DBA-EN (Cross-Sector)	DBA-EN-SF (this document)	v0.9 — DRAFT	DOE / FERC / NERC / PHMSA / TSA	DBA-EN-UC-1 (v0.1), DBA-EN-UC-2 (v0.1), DBA-VC (v0.3, cross-series)

Section 2 — Energy Sector Taxonomy

2.1 Organizing Framework

The Energy sector’s facility taxonomy is organized into two primary branches corresponding to the NIPP Energy sector structure: Electricity and Oil and Natural Gas. The Electricity branch covers generation (GenCo), transmission (TransCo), distribution (DisCo), and the control-company function (CntlCo). The Oil and Natural Gas branch covers crude oil production, midstream, and petroleum processing; and natural gas production, transmission, distribution, LNG, and storage.

The taxonomy in this section is the authoritative reference for series-wide facility scoping decisions. When a new facility class is proposed for any sub-sector series, its placement — which sub-sector series it belongs to, which consequence chain governs its inclusion, and which regulatory anchor applies — is determined against this taxonomy. The governing principle is consequence-chain placement: a facility belongs in the sub-sector series whose primary consequence chain it affects most directly.

2.2 GenCo — Generation by Type

The generation taxonomy spans seven primary generation types. Nuclear generation is addressed in the DBA-MA nuclear series and is referenced here for completeness; its grid consequence chain (Zaporizhzhia: loss of Europe’s largest nuclear plant as both a nuclear safety event and a grid reliability event) is addressed in the DBA-ES series and the DBA-MA Introduction.

Generation Type	Sub-Types	Primary Regulators	DBA-EN Series Treatment	FC Status
Fossil fuel	Coal, natural gas, oil-fired	TSA (security); NERC/FERC (reliability); State environmental (emissions)	Gas-fired generation is addressed through a gas-electric cascade in DBA-ES-GC-FC4 and DBA-GAS-FC1. Coal: DBA-ES-GN-FC6. Oil-fired: DBA-ES-GN-FC7.	FC4 (gas), FC6 (coal), FC7 (oil-fired)
Nuclear	Large LWR (PWR, BWR), SMR, VVER	NRC (US); STUK (Finland); SNRIU (Ukraine); IAEA (international)	Addressed in the DBA-MA nuclear series. Grid consequence chain (Zaporizhzhia) addressed in DBA-ES and DBA-MA Introduction §6.3.	DBA-MA series
Hydropowe	Large storage	FERC Part 12;	Large hydroelectric	DBA-ES-HY-

r	(>500 MW), run-of-river, pumped storage, tidal	NERC CIP-014; USCOLD; Army Corps of Engineers	storage is addressed in DBA-ES-HY-FC1. Run-of-river and pumped storage: deferred FC. Tidal: not yet scoped.	FC1
Solar	Utility-scale PV, solar thermal (CSP), distributed rooftop	FERC; NERC (interconnection); State PUC (distributed)	Utility-scale solar at cluster level: deferred FC. Distributed solar is addressed as an edge generation component in DBA-ES-DI-FC5.	Partial (FC5)
Wind	Onshore utility-scale, offshore, distributed	FERC; NERC; BOEM (offshore); State PUC	Wind at cluster level: deferred FC. Distributed wind is addressed as an edge generation component in DBA-ES-DI-FC5. Offshore wind: DBA- ES-GN-FC8.	Partial (FC5, FC8)
Battery / BESS	Utility-scale BESS, distributed storage	FERC Order 841; NERC; State PUC	Utility-scale BESS: DBA-ES-GN-FC8. Distributed BESS is addressed as the edge generation component in DBA-ES-DI-FC5.	FC8, partial (FC5)
Geothermal, biomass, wave, tidal	Various	FERC; State PUC; applicable environmental agencies	Not yet scoped. Consequence chains have not yet been analyzed.	Not yet

2.3 TransCo — Transmission Owners and Operators

The transmission taxonomy covers the physical infrastructure connecting generation to load at bulk power system voltage levels. Voltage class specification and bus bar identifiers per circuit are the technical parameters that govern facility class placement within DBA-ES: the 345 kV threshold separates FC2 (EHV substations) from sub-transmission facilities outside the current series scope.

Infrastructure Element	Technical Specification	Regulatory Anchor	DBA-EN Series Treatment
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EHV transmission substations	345–765 kV; metropolitan ring configuration; CIP-014 Tier 1/2 designation; LPT with RT_HVLLC $\geq$ 12 months	NERC CIP-014; FERC Order 693; TPL standards	DBA-ES-TX-FC2 (v0.3)
HVDC interconnects	>2,000 MW; cross-interconnection role; LCC or VSC technology; point-to-point (primary scope) or multi-terminal (deferred)	FERC; NERC FAC standards; applicable TPL	DBA-ES-TX-FC3 (v0.3)
Transmission lines	Voltage class per facility: 115 kV, 230 kV, 345 kV, 500 kV, 765 kV. Bus bar identifiers per circuit.	NERC FAC-001/002; FERC; State PUC	Deferred FC; addressed at the system level in TPL standards
SCADA and control communications	ICCP; SCADA RTUs; substation automation; protective relaying	NERC CIP-005/006/007; FERC Order 706	CE events in FC2, FC3; SL events in CntlCo instrument
Protective relay systems	Distance, differential, overcurrent, frequency, voltage protection; PRC-024 ride-through settings	NERC PRC standards; IEEE relay standards	CE-2/CE-3 events in FC2; CE-3 in FC5; UFLS position paper

2.4 DisCo — Distribution Functions

The distribution taxonomy covers the infrastructure delivering power from the transmission system to end-use customers. The PMU (Phasor Measurement Unit) is specifically called out in the DisCo control systems because it is the primary measurement instrument underlying the UFLS frequency wavefront analysis in the Predictive UFLS Ride-Through Position Paper — the cross-reference between DisCo instrumentation and the UFLS paper is a series linkage that must be maintained.

Infrastructure Element	Technical Specification	Regulatory Anchor	DBA-EN Series Treatment
Distribution	HV/EHV input (115	NERC CIP (where	DBA-ES-DI-FC5 (v0.4)

substations with edge generation	kV+); 69–110 kV output; microgrid or DER assets; islanding capability with Command Broker architecture	applicable); IEEE 1547; State PUC; FERC Order 2222	
Loads	Commercial, industrial, residential. Load classification: firm/fixed vs. sheddable. Defense-critical loads: DSB finding (31 of 34 DoD most critical assets on commercial DisCo infrastructure).	State PUC; FERC (wholesale); DOD (defense-critical)	XC events in FC2, FC5; Tier 3 acceptance criteria
Distribution control systems	SCADA (2–5 second polling cycle); DCS; PLC; EMS (Energy Management System); PMU (Phasor Measurement Unit — wavefront analysis; cross-reference Predictive UFLS Ride-Through Position Paper)	NERC CIP; State PUC; IEEE standards	CE events in FC5; PMU data cited in UFLS position paper
Electricity markets	Wholesale market dispatch; retail rate structures; demand response programs; ancillary services	FERC (wholesale); State PUC (retail)	Market timing dependency in XC-4 events (FC3, FC4)

2.5 CntlCo — Balancing Authority, Reliability Coordinator, Transmission Operator

The CntlCo taxonomy is more tool-specific than the physical infrastructure taxonomies because the CntlCo’s design basis is anchored in its software-data layer as much as its physical and human layers. The RC and BA tool suites, the NERC CIP compliance registry classification scheme, and the primary and backup control center communications architecture are all design basis elements addressed in the DBA-ES-CntlCo functional class instrument.

Function / Tool	Description	NERC Registration	DBA-EN Series Treatment
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Reliability Coordinator (RC) tools	SCADA; Electronic Tagging System; Telephone System; OATI Suite of Applications; State Estimator; EPRI TagNet; Power World; RCIS (Reliability Coordinator Information System)	RC	SL events in DBA-ES-CntlCo §5; RC extension §7.2; SABD metric governs state estimator loss consequence
Balancing Authority (BA) tools	Transmission Energy Scheduler; Eastern Interconnect Frequency Monitor; Security Constrained Unit Commitment; Telephone System; SCADA; RCIS; Energy Management System (EMS)	BA	SL events in DBA-ES-CntlCo §5; BA primary case §2; SABD governing metric
NERC CIP compliance registry	Critical cyber asset classification: BCA (BES Cyber Asset); EACMS (Electronic Access Control and Monitoring System); PACS (Physical Access Control System); PCA (Protected Cyber Asset). Impact classification: High / Medium / Low.	All CIP-registered entities	CE-4 (cyber intrusion) in FC2; SL-5 in CntlCo; governs CIP-014 designation trigger for FC2 scope
Primary and backup control center communications	Data communications; Voice; Radio; Telephone System; SCADA; ICCP. Primary and BCC must be geographically independent per DBA-ES-CntlCo acceptance criteria.	BA, RC, TOP	PL-4, PL-5 events in DBA-ES-CntlCo; BCC geographic independence acceptance criterion
Transmission Operator (TOP)	Operates transmission facilities in real time; implements RC and BA directives; controls switching at FC2 and FC3 facilities; must maintain	TOP	TOP extension in DBA-ES-CntlCo §7.3

	manual field switching capability under SCADA loss		
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2.6 Oil and Natural Gas

The Oil and Natural Gas branch of the energy taxonomy covers two commodity streams with partially overlapping infrastructure. The governing cross-sector cascade — the gas-electric bidirectional feedback loop — is addressed in Section 3. The taxonomy below covers the upstream, midstream, downstream, and storage segments of each commodity stream.

Segment	Infrastructure	Technical / Operational Notes	Primary Regulators	DBA-EN Series Treatment
Crude Oil — Upstream	Onshore fields; offshore fields; gathering systems	Wellhead pressure management, produced water handling, and gathering line integrity	PHMSA; EPA; State environmental; BSEE (offshore)	DBA-OIL-FC2 (v0.1) for offshore production. Onshore upstream: not yet scoped.
Crude Oil — Midstream	Terminals; pipelines; storage (tank farms)	Crude transport by pipeline, rail, and marine. Tank farm fire and explosion risk. Colonial Pipeline paradigm applicable.	PHMSA; TSA; FERC; EPA	DBA-OIL-FC3 (v0.1) for terminals and storage. DBA-OIL-FC4 (v0.3) for transport pipelines. SDD metric governs transport consequence.
Petroleum Processing	Refineries, blending terminals, petrochemical plants	Process unit fire and explosion as the governing SE event. Long lead time for major process units. RT_HVLLC concept applicable.	TSA; EPA; OSHA; State environmental	DBA-OIL-FC1 (v1.3)
Natural Gas — Upstream	Onshore fields; offshore fields; gathering systems; processing plants	Gas processing separates NGLs from methane. Processing plant disruption reduces the pipeline quality	PHMSA; FERC; EPA; BSEE (offshore)	Not yet scoped in DBA-GAS series

		gas supply.		
Natural Gas — Transmission	Interstate and intrastate pipelines; compressor stations; metering and regulation (M&R) stations	Compressor stations in critical GenCo supply corridors are the governing FC4 facility class. GSD metric governs electric consequence.	FERC; PHMSA; TSA (security directives)	DBA-GAS-FC1 (v1.6); DBA-ES-GC-FC4 (v0.5)
Natural Gas — Distribution	Local distribution companies (LDCs), service laterals, and customer meters	Residential heating dependency: dual-use pipeline (generation fuel + heating fuel) produces compound XC-4 consequence during winter peak.	State PUC; PHMSA	DBA-GAS-FC4 (v0.1); XC-4 in DBA-ES-GC-FC4
LNG Facilities	Liquefaction terminals; storage tanks; regasification terminals; import/export terminals	LNG as a backup supply for pipeline disruption. Sabotage of LNG import terminal removes backup gas supply during extended pipeline outage.	FERC; PHMSA; USCG; DOE (export)	DBA-GAS-FC-LNG (annex DBA-GAS-UC-LNG v1.1)
Storage	Underground storage (depleted reservoirs, aquifer, salt cavern); above-ground LNG storage	Strategic storage buffer: days to weeks of deliverability. Storage disruption extends GSD duration beyond compressor station recovery.	FERC; PHMSA; State	DBA-GAS-FC3 (v0.1)

Section 3 — The Gas-Electric Cascade: Cross-Sub-Sector Postulated Event

3.1 Status as a First-Class Postulated Event

The gas-electric bidirectional cascade is established in this document as a first-class cross-sub-sector postulated event at the DBA-EN level. It is not a secondary consequence or a beyond-design-basis contingency: it is a primary postulated event that must be analyzed in both the DBA-ES series (from the electric system’s perspective) and the DBA-GAS series (from the gas system’s perspective). The two analyses are complementary; neither substitutes for the other.

The bidirectional character of the cascade — gas disruption reduces electric generation; electric disruption reduces gas compression — was established in DBA-ES-Sector_Framework §3.5 and is carried through every relevant FC instrument. Winter Storm Uri (February 2021) demonstrated both directions simultaneously and empirically. An adversary who understands the feedback structure can initiate the cascade from whichever end is more accessible and rely on the bidirectional dependency to amplify the consequence without further action.

3.2 Cascade Analysis Table

Cascade Direction	Mechanism	Governing Metric	Primary DBA-EN Instrument
Gas → Electricity	Natural gas compressor station loss reduces pipeline delivery pressure below the GenCo minimum inlet threshold. Gas-fired generation curtails output or trips offline. The electric system loses generation capacity. Approximately 40% of US electric generation is gas-fired; gas transmission disruption in a critical corridor reduces generation capacity at the electric system level.	GSD (Generation Supply Duration): continuous period of full gas-fired generation curtailment due to insufficient pipeline pressure. Defined in DBA-ES-GC-FC4 §1.3.	DBA-ES-GC-FC4; DBA-GAS-FC1
Electricity → Gas	Electric supply loss to electrically driven compressor stations reduces compression capacity. Pipeline pressure in the supply corridor declines. Gas supply to gas-fired generation and heating customers is curtailed. Self-reinforcing: reduced gas supply further reduces gas-fired generation, further reducing	GSD (reverse cascade): same metric applied to the electrically-initiated cascade. TCI (Transfer Capability Isolation, from DBA-ES-TX-FC3) may apply if	DBA-ES-GC-FC4 (XC-2); DBA-ES-TX-FC2; DBA-ES-TX-FC3

	electric supply to compressor station auxiliaries.	HVDC or transmission loss initiates the electric supply disruption.	
Compound (bidirectional)	Winter Storm Uri (February 2021) demonstrated the full bidirectional cascade: extreme cold simultaneously increased electric demand and reduced gas-fired generation supply by disabling compressor stations and gas wells. The compound event eliminated both directions of the cascade mitigation simultaneously. An adversary can replicate this compound cascade deliberately through coordinated timing.	Compound GSD + load curtailment. XC-4 event in DBA-ES-GC-FC4: winter peak demand timing maximizes both heating fuel and electric generation consequences.	DBA-ES-GC-FC4 (XC-4); cross-reference DBA-GAS-FC1

3.3 Cross-Sub-Sector Coordination Requirement

The gas-electric cascade is the primary reason DBA-EN-SF exists as a document distinct from the sub-sector frameworks. Each sub-sector framework addresses the cascade from its own consequence perspective: DBA-ES instruments analyze the electric consequence of gas infrastructure failure; DBA-GAS instruments analyze the gas infrastructure consequence of electric supply failure. The DBA-EN-SF level is where the full bidirectional cascade is formally established as a postulated event category that both sub-sector analyses must address.

3.4 Cross-Sector Consequence Elevation Rule

The NERC gap this rule addresses. The North American Electric Reliability Corporation’s CIP standards assign impact levels to bulk electric system cyber assets based on the electric-sector consequences of their loss. NERC CIP does not require, and has not established, any mechanism by which the consequence designation of a facility in another sector elevates the design basis or impact classification of the energy sector asset that supplies it. The DBA-EN-SF cross-sector consequence elevation rule fills this gap as a pre-regulatory design basis requirement.

The rule. A DBA-EN series energy sector asset whose loss would deprive a facility carrying a Tier 2 or Tier 3 consequence designation in any DBA series instrument of its primary energy supply must carry tightened acceptance criteria for that supply relationship in its own DBA-EN design basis.

Trigger condition. The rule fires when two conditions are simultaneously satisfied. First, the dependent facility carries a Tier 2 or Tier 3 consequence designation in a DBA series instrument. Second, the energy sector asset is the sole necessary supply path: the dependent facility does not

have an alternate energy supply of equivalent capacity. This single-path dependency criterion is drawn directly from the FC4 facility class definition (DBA-ES-GC-FC4 §1.2).

Self-referential trigger and series growth. The trigger is self-referential within the DBA series: the Tier 2 or Tier 3 designation of the dependent facility is established by its own DBA instrument, not by any external regulatory designation. This means the rule automatically expands as the DBA series matures. When the DBA-W series assigns a Tier 3 designation to a water treatment facility, every DBA-EN energy asset that is the sole supply path for that facility is immediately subject to the elevation rule.

Bidirectional implementation. The rule is implemented bidirectionally. The forward direction — the energy sector carries the design basis obligation imposed by dependent sectors' consequence tiers — is the foundational implementation established at v0.3. The reverse direction is now established for the named dependent series in the trigger table below. The European EPCIP / NIS2 framework is the governing international analog.

Named-series elevation trigger table:

Series	Triggering Tiers	Direction	Governing Instrument and Notes
DBA-W (Water Sector)	Tier 2 and Tier 3	Bidirectional. Forward: DBA-EN energy asset is the sole electric supply path for a Tier 2 or Tier 3 DBA-W facility. Reverse: DBA-W facility is the sole cooling water supply path for a Tier 2 or Tier 3 DBA-ES thermoelectric facility.	DBA-EN-SF §5.4 (DBA-EN-SF / DBA-W Boundary Protocol). Full boundary protocol established at v0.4. Life-safety cascade anchored by dialysis center dependency (DBA-W §8.2)—instrument-level incorporation in DBA-W FC instruments: pending DBA-W series next revision.
DBA-DC (Data Center Series)	Tier 2 and Tier 3	Forward only. DBA-DC facilities are electricity consumers; no energy sector reverse dependency exists.	DBA-DC-L1 §5.2 and §5.3 establish the Tier 2 (grid-level frequency and voltage consequence) and Tier 3 (foundational digital dependency cascade) designations that trigger elevation. Instrument-level incorporation of the elevation obligation in DBA-DC FC instruments is an open item (OI-DC-1).

DBA-MA Nuclear Series	Tier 2 and Tier 3	Bidirectional. Forward: DBA- EN energy asset is the sole off-site power supply path for a nuclear facility; elevation implements the Station Blackout design basis (10 CFR 50.63). Reverse: nuclear facility loss is a Tier 2 or Tier 3 DBA-ES generation event.	Forward direction: NUC-001-4 (Nuclear Plant Interface Coordination) is the NERC implementing instrument. Instrument- level incorporation of the elevation obligation in DBA-MA nuclear FC instruments is an open item (OI-MA-1).
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Notification protocol. When a DBA-EN series instrument is revised in a way that changes the consequence tier of a supply relationship governed by this rule, the revision history of that instrument must identify which cross-series instruments carry a dependency on that designation. Those instruments are flagged for review at their next scheduled revision.

Open items — instrument-level incorporation in dependent series. OI-DC-1: DBA-DC series instrument-level incorporation of the forward-direction elevation obligation. Resolution: DBA-DC-L1 next substantive revision. OI-MA-1: DBA-MA nuclear series instrument-level incorporation of the bidirectional elevation obligation. Resolution: DBA-MA US Regulatory Case next substantive revision.

Section 4 — Series-Wide Architectural Requirements

4.1 DB → DBA → DBA-MA Three-Layer Progression

All instruments in the Energy sector series — sub-sector frameworks, facility-class instruments, functional class instruments, and use case documents — apply the DB → DBA → DBA-MA three-layer progression. The progression is non-negotiable: a facility-class instrument that addresses only the DBA layer without explicitly establishing the DB layer, or that addresses the DBA-MA extension without the DBA envelope, is architecturally incomplete.

Non-nuclear sector frameworks and facility-class instruments cannot assume reader familiarity with the DB → DBA → DBA-MA architecture and must state it explicitly. This requirement was established in DBA-ES-Sector_Framework and is adopted here as an Energy-sector-wide requirement.

4.2 Dual Design Basis Architecture

Consistent with DBA-ES §3.4 and the AI Governance series (Project 2), any energy sector facility may simultaneously carry a sector-specific physical Design Basis established by the applicable DBA-EN series instrument and an AI Governance Design Basis governing the deployment of ML and Gen AI systems in the facility’s operational and control environment. The two design bases interact at the DBA level and must be developed in coordination.

Each sub-sector framework and facility-class instrument must acknowledge the dual-DB architecture and identify the interaction points between the physical DBA and the AI Governance DBA specific to its facility type. This requirement applies to all new instruments and to existing instruments on their next substantive revision pass.

4.3 Bright Line and Command Broker Architecture

The Bright Line and Command Broker architecture is a series-wide requirement for all energy sector instruments. The Bright Line is the Bright Line as defined in FD-BL (Foundational Definition — The Bright Line): a threshold on individual actions, not a property of a component, a technology class, or an architectural layer. Under FD-BL §4.2–§4.3, an action is above the line when its incorrect performance could produce a consequence at or above the facility’s severity threshold and requires substantive, architecturally enforced authorization by a qualified Command Broker (FD-BL §4.5); an action is below the line when it could not. Whether a component is deterministic or generative does not decide placement — determinism is the scope carve-out of FD-BL §4.4, under which a component that is deterministic within a stated input bound and formally testable as deployed is outside the Command Broker obligation. Energy sector instruments do not restate this definition; they inherit it by reference to FD-BL and apply it to their facility types. Each sub-sector framework — DBA-ES, DBA-GAS, and DBA-OIL — and every facility-class and functional-class instrument beneath them takes the Bright Line as defined in FD-BL and develops only the sector-specific placement and Command Broker application for its facilities.

The term “Bright Line” is used exclusively in the FD-BL sense throughout all DBA-EN series instruments. Any other use of the term in an energy sector instrument — for example, to mean federal coordination triggers or escalation thresholds — must be replaced with a distinct term. The applicable replacement term for federal coordination thresholds is “Federal Coordination Threshold.”

4.4 Consequence Metrics

Six consequence-duration metrics are established across the Energy sector series. Each is a series standard and must be used consistently in any instrument that addresses the relevant consequence type.

RT_HVLLC (Recovery Timeline for High-Value Long-Lead Components): the time required to restore a facility to service following destruction of its most difficult-to-replace major component, expressed in months. Established as a series standard in DBA-ES-Sector_Framework §6.2. Originated in GenMix Optimization Whitepaper (Roxey, Eclectic Technologies, 2026). Applies to all FC instruments where long-lead-time equipment is the governing consequence element.

TCI (Transfer Capability Isolation): the continuous duration during which cross-interconnection transfer capability is unavailable following loss or block of an HVDC link, expressed in hours, days, or months. Established in DBA-ES-TX-FC3 §1.3. Applies to HVDC facility class instruments.

GSD (Generation Supply Duration): the continuous period during which gas-fired generation capacity in an affected GenCo supply corridor is curtailed to zero due to insufficient pipeline delivery pressure, expressed in hours or days. Established in DBA-ES-GC-FC4 §1.3. Applies to gas compressor stations and gas transmission instruments where the governing consequence chain runs through the gas-electric cascade.

PPSD (Pipeline Pressure Supply Duration): the continuous period during which pipeline delivery pressure at the transmission system level is insufficient to maintain pipeline-quality gas supply to downstream compressor stations and GenCo facilities, following a major gas processing plant outage, expressed in hours or days. Established in DBA-GAS-FC2 §1.3 as the governing consequence metric for natural gas processing plant outage. PPSD is upstream of and sequential to GSD: PPSD measures the consequence of processing plant loss on pipeline pressure supply duration; GSD measures the downstream consequence of that pipeline pressure loss on gas-fired generation supply duration. The PPSD → GSD sequential relationship is formally established in DBA-EN-UC-1 §5.1. Applies to gas processing facility class instruments and to any cross-sector cascade analysis where processing plant loss is the upstream initiating condition.

SDD (Supply Delivery Duration): the continuous interval of pipeline delivery curtailment from the initiating event to restoration of full throughput capacity at the governing delivery point, expressed in hours or days. Established in DBA-OIL-FC4 v0.3 §§1.3 and 4.2 as the governing consequence metric for the crude and product pipeline transport facility class. SDD is measured at the governing delivery point — the terminal, refinery intake, or distribution hub whose curtailment produces the highest-tier consequence — not at the point of physical damage. SDD is not additive across pipeline segments: where multiple segments are affected, the governing SDD is the longest single-segment critical path, not the cumulative sum across segments. A Zone 1/2/3 acceptance criterion framework with T₁/T₂ calibration thresholds governs consequence

tier assignment; site-specific calibration is required at implementation. Applies to crude and product pipeline transport facility class instruments and to cross-sector cascade analysis where liquid fuel delivery curtailment is the governing upstream consequence.

SABD (Situational Awareness Blackout Duration): the continuous interval between loss of state estimator validity and restoration of coordinated real-time operational authority at the Balancing Authority, Reliability Coordinator, or Transmission Operator level, expressed in minutes or hours and established in DBA-ES-CntlCo v0.2 §1.2 as the governing consequence metric for the CntlCo functional class. SABD begins at the moment the state estimator is confirmed non-convergent, produces outputs flagged as unreliable, or is taken out of service due to suspected data integrity compromise. SABD ends at the moment a qualified system operator confirms the state estimator is producing convergent, validated results and resumes issuing dispatch instructions based on its output. Backup control center (BCC) transition does not automatically terminate SABD: a successful BCC transition that does not restore state estimator convergence leaves the SABD clock running at the BCC. A Zone 1/2/3 escalation structure with T₁/T₂ calibration thresholds governs consequence tier assignment; site-specific calibration is required. Applies to CntlCo functional class instruments and to any cross-sector cascade analysis where situational awareness loss is a contributing condition.

4.5 Facility Class Naming Convention

The naming convention for Energy sector facility class instruments is: DBA-[SUB-SECTOR]-[TYPE CODE]-FC[NUMBER]. Type codes are assigned to distinguish facility classes within the same sub-sector by their governing infrastructure type. The DBA-DC series retains its existing designation without modification. Facility classes promoted from an existing class take the next sequential FC number (for example, DBA-OIL-FC5, carved from DBA-OIL-FC3); facility classes promoted from a named standalone use case may instead retain the use-case mnemonic in place of a sequential number (for example, DBA-GAS-FC-LNG, promoted from DBA-GAS-UC-LNG), preserving traceability to the originating instrument.

A sub-sector may further organize its facility classes into named subsector groupings — for example, the Critical Loads grouping (DBA-ES-CL) within the Electric sub-sector, which groups load-side facility classes whose design basis is electric-sector-dominated and whose compromise elevates across sectors. Such groupings are organizational and do not alter the designation of the instruments they contain.

Designation	Sub-Sector	Type Code	Facility / Functional Class
DBA-EN-SF	Energy (parent)	SF = Sector Framework	This document — Energy sector parent
DBA-ES-Sector_Framework	Electric	SF = Sector Framework	Electric sub-sector parent

DBA-ES-HY-FC1	Electric	HY = Hydroelectric	Large hydroelectric storage (>500 MW)
DBA-ES-TX-FC2	Electric	TX = Transmission	Large EHV transmission substation (345–765 kV)
DBA-ES-TX-FC3	Electric	TX = Transmission	HVDC interconnect (>2,000 MW)
DBA-ES-GC-FC4	Electric	GC = Gas-electric Cascade	Major natural gas compressor station (GenCo supply corridor)
DBA-ES-DI-FC5	Electric	DI = Distribution	Distribution substation with edge generation
DBA-ES-CntlCo	Electric	CntlCo = functional class	BA / RC / TOP functional class instrument
DBA-ES-GN-FC6	Electric	GN = Generation	Coal-fired generation (>750 MW or must-run)
DBA-ES-GN-FC7	Electric	GN = Generation	Oil-fired generation and peakers (>100 MW or RMR)
DBA-ES-GN-FC8	Electric	GN = Generation	Renewable generation and BESS
DBA-DC (unchanged)	Electric / IT	DC = Data Center	Hyperscale data center (>100 MW IT load)
DBA-GAS-Sector_Framework	Natural Gas	SF = Sector Framework	Natural gas sub-sector parent
DBA-GAS-FC1	Natural Gas	—	Natural gas transmission
DBA-OIL-Sector_Framework	Petroleum	SF = Sector Framework	Petroleum sub-sector parent
DBA-OIL-FC1	Petroleum	—	Petroleum refining
DBA-OIL-FC4	Petroleum	—	Crude and product transport — trunk and product pipelines (SDD metric)
DBA-GAS-FC-LNG	Natural Gas	LNG (mnemonic)	LNG terminals — liquefaction, storage, and marine transfer (promoted from

			DBA-GAS-UC-LNG)
DBA-OIL-FC5	Petroleum	—	Marine crude and product terminals — strategic import/export (carved from FC3)

Section 5 — Cross-Sector Dependencies Beyond Gas-Electric

5.1 Water Sector Dependency

The water-energy interdependency is bidirectional. The forward direction runs from the energy sector to the water sector: water treatment plants, pumping stations, and distribution infrastructure are significant electric loads whose continuous operation depends on grid supply. Loss of electric supply to a major water utility is a Tier 3 consequence in the DBA-ES series, because the downstream life-safety cascade — loss of potable water supply to hospitals, dialysis centers, and dense residential populations — constitutes a cross-sector consequence chain of the severity that defines Tier 3. DBA-W §8.2 establishes dialysis centers as the most time-critical dependent facility class.

The reverse direction runs from the water sector to the energy sector: thermoelectric generating facilities — nuclear, natural gas, and coal — withdraw large volumes of water for cooling. A sustained water supply disruption affecting a power generation facility’s cooling water intake can force the facility offline, compounding any grid reliability event. The formal governance boundary for this interdependency is established in Section 5.4.

5.2 Data Center Dependency

Hyperscale data centers are addressed in the DBA-DC series, which is an Electric sector sub-series instrument. Data centers are electricity consumers — not electricity infrastructure — per NIPP taxonomy. Their classification in the Electric sector design basis series reflects their consequence chain, not their sector membership. The WM-3 workload migration cross-interconnection load transient is the governing cross-sector cascade event for the data center class and is addressed in DBA-DC-L1 and DBA-ES-Sector_Framework.

The data center series is the founding member of the Electric sector’s Critical Loads subsector (DBA-ES-CL), defined in DBA-ES §8.5. The forward-only elevation relationship described above is the first instance of a subsector-wide pattern rather than a property unique to data centers; subsequent Critical Loads members may carry a reverse energy-sector dependency.

5.3 Nuclear Sector Grid Dependency

Nuclear generating facilities are significant contributors to the bulk electric system. The Zaporizhzhia scenario demonstrated that military action against a nuclear facility is simultaneously a nuclear safety event and a grid reliability event. The DBA-MA nuclear series addresses the nuclear safety consequences; the DBA-ES series addresses the grid reliability consequences of nuclear facility loss. The boundary is defined in the DBA-MA Introduction §6.3. The NERC regulatory instrument governing the interface between nuclear plant operations and the bulk electric system is NUC-001-4 (Nuclear Plant Interface Coordination). NUC-001-4’s NPIR coordination mechanism is the existing regulatory pathway through which nuclear-facility-

specific grid reliability requirements are made binding on Transmission Operators, Balancing Authorities, and Reliability Coordinators.

5.4 DBA-EN-SF / DBA-W Boundary Protocol

This section establishes the formal governance boundary between DBA-EN-SF and DBA-W for the water-energy interdependency. The boundary protocol addresses three questions: (a) instrument assignment for the energy supply consequence chain; (b) cross-sector consequence elevation obligation; and (c) the inheritance obligation running from DBA-EN-SF to the DBA-ES and DBA-W series.

Instrument assignment — energy supply consequence chain. A DBA-ES facility-class instrument governs the energy supply consequence chain at any facility whose loss deprives water sector facilities of their primary electric power supply. The DBA-ES instrument for the relevant facility class carries the Design Basis obligation for the electric supply consequence, including the Tier 3 designation for water sector life-safety cascades. The DBA-W series governs the water sector consequence chain at facilities within its own series scope.

Cross-sector consequence elevation obligation — forward direction (energy to water). A DBA-EN energy asset that is the sole electric supply path for a water sector facility carrying a Tier 2 or Tier 3 consequence designation in the DBA-W series must carry tightened acceptance criteria for that supply relationship, consistent with the Cross-Sector Consequence Elevation Rule in §3.4.

Cross-sector consequence elevation obligation — reverse direction (water to energy). A DBA-W water sector facility that is the sole cooling water supply path for a thermoelectric generation facility carrying a Tier 2 or Tier 3 consequence designation in the DBA-ES series must carry tightened acceptance criteria for that supply relationship in the DBA-W instrument governing the relevant facility class.

Inheritance obligation. The DBA-ES series inherits from this section the obligation to identify water sector cross-sector dependencies in each facility-class instrument and to apply the forward-direction elevation rule. The DBA-W series inherits the obligation to apply the reverse-direction elevation rule. Until the reverse-direction obligation is incorporated into each DBA-W facility-class instrument, each DBA-ES FC instrument must note the water sector cross-sector dependency as an open item in its gaps and recommendations section, consistent with DBA-ES-HY-FC1 §4.5 and DBA-ES-TX-FC2 §4.5.

Section 6 — Development Status and Open Items

6.1 Current Development Status

The Electric sub-sector series (DBA-ES) is the most developed of the three sub-sector series. DBA-ES-Sector_Framework is at v0.10. Eleven facility-class instruments, one functional-class instrument, and one sector-level MA companion are active: DBA-ES-HY-FC1 (v0.5), DBA-ES-TX-FC2 (v0.3), DBA-ES-TX-FC3 (v0.3), DBA-ES-GC-FC4 (v0.5), DBA-ES-DI-FC5 (v0.4), DBA-ES-CntlCo (v0.3), DBA-ES-GN-FC6 (v0.2), DBA-ES-GN-FC7 (v0.2), DBA-ES-GN-FC8 (v0.2), DBA-DC (v1.9), DBA-ES-POL (v0.1), DBA-ES-MA-SF (v0.2), and DBA-ES-MA-QB (v0.1). All FC instruments carry the full DB → DBA → DBA-MA three-layer structure, including the MA applicability four-filter assessment and CB-1 through CB-5 communications design basis.

The Natural Gas sub-sector series (DBA-GAS) is at v1.7 for its sector framework with eight active instruments: DBA-GAS-FC1 (v1.6), DBA-GAS-FC2 (v0.2), DBA-GAS-FC3 (v0.1), DBA-GAS-FC4 (v0.1), DBA-GAS-FC-LNG (promoted; annex DBA-GAS-UC-LNG v1.1), DBA-GAS-POL (v1.3), and DBA-GAS-MA-SF (v0.2). The series has completed the full three-layer architecture for the transmission and processing classes; LNG terminals have been promoted from use-case status to a standing facility class (DBA-GAS-FC-LNG).

The Petroleum sub-sector series (DBA-OIL) is at v1.8 for its sector framework with six active instruments: DBA-OIL-FC1 (v1.3), DBA-OIL-FC2 (v0.1), DBA-OIL-FC3 (v0.1), DBA-OIL-FC4 (v0.3), DBA-OIL-FC5 (promoted), and DBA-OIL-MA-SF (v0.1). The SDD metric is now established at both the FC4 level and sanctioned at this parent document level in §4.4. Strategic marine terminals have been carved from FC3 and registered as a standing facility class (DBA-OIL-FC5).

6.2 Open Items

OI-1: CLOSED. DBA-ES-GC-FC4 / DBA-GAS-FC1 scope boundary coordination. Resolved by §3.4.

OI-2: CLOSED. GenCo facility classes for coal, oil-fired, and renewable generation types. Resolved by DBA-ES-GN-FC6, FC7, FC8 v0.1 issuance.

OI-3: Partially addressed. Upstream and midstream petroleum facility classes. DBA-OIL-FC2 (offshore production), DBA-OIL-FC3 (terminals and storage), and DBA-OIL-FC4 (transport pipelines) are now at first issue. Onshore upstream production has not yet been scoped.

OI-4: Partially addressed. Natural gas distribution and storage facility classes. DBA-GAS-FC3 (underground storage, v0.1) and DBA-GAS-FC4 (local distribution, v0.1) are now at first issue. The residential heating dependency is addressed from the gas sector perspective.

OI-5: CLOSED. DBA-EN-SF / DBA-W boundary. Established in Section 5.4.

OI-6: CLOSED. DBA-EN-UC-2 registered and issued as v0.1.

Section 7 — References

Regulatory and Policy References

Department of Homeland Security. National Infrastructure Protection Plan 2013: Partnering for Critical Infrastructure Security and Resilience. Washington, D.C.: DHS, 2013.

Presidential Policy Directive 21 (PPD-21): Critical Infrastructure Security and Resilience. Washington, D.C.: The White House, February 2013.

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. NUC-001-4: Nuclear Plant Interface Coordination. Atlanta: NERC. Current version. Governs coordination between Nuclear Plant Generator Operators and Transmission Entities for the development and maintenance of Nuclear Plant Interface Requirements (NPIRs). Cited in §5.3 (Nuclear Sector Grid Dependency) as the NERC regulatory instrument for the nuclear-grid operational interface.

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Pipeline and Hazardous Materials Safety Administration. 49 CFR Part 192: Transportation of Natural and Other Gas by Pipeline. Washington, D.C.: PHMSA. Current version.

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DBA-EN Series Instruments

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.9 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-HY-FC1: Large Hydroelectric Storage Facilities. Version 0.4 — DRAFT. Eclectic Technologies, May 2026.

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-TX-FC3: HVDC Interconnect. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GC-FC4: Natural Gas Compressor Station. Version 0.3 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-DI-FC5: Distribution Substation with Edge Generation. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-CntlCo: Electric Sector Control-Company Functional Class. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GN-FC6: Coal-Fired Generation. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GN-FC7: Oil-Fired Generation. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-GN-FC8: Renewable Generation and BESS. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-GAS-Sector_Framework. Version 1.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-GAS-FC1: Natural Gas Transmission. Version 1.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-GAS-FC2: Natural Gas Processing. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-GAS-FC3: Underground Storage. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-GAS-FC4: Local Distribution. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-OIL-Sector_Framework. Version 1.7 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-OIL-FC1: Petroleum Refining. Version 1.3 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-OIL-FC2: Offshore Crude Production. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-OIL-FC3: Crude and Product Terminals and Storage. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-OIL-FC4: Crude and Product Transport. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Supporting Documents

Roxey, Tim. GenMix Optimization Whitepaper. Version 0.3 — DRAFT. Eclectic Technologies, 2026. (Origin of RT_HVLLC construct.)

Roxey, Tim. Predictive UFLS Ride-Through Position Paper. Version 0.3 — DRAFT. Eclectic Technologies, 2026. (PMU wavefront analysis; cross-reference DisCo PMU taxonomy entry.)

Defense Science Board Task Force on DoD Energy Strategy. “More Fight — Less Fuel.” Washington, D.C.: Office of the Under Secretary of Defense for Acquisition, Technology and Logistics, February 2008.

Revision History

Version	Date	Author	Description
0.1	May 2026	T. Roxey	Initial working draft. Established Energy sector parent document architecture. Defined three sub-sector series: DBA-ES (Electric), DBA-GAS (Natural Gas), DBA-OIL (Petroleum). Established an inductive architecture principle. Named the gas-electric bidirectional cascade as the governing cross-sub-sector interdependency. Confirmed DBA-ES retains its own designation and is not renamed under a DBA-EN hierarchy.
0.2	June 2026	T. Roxey	Substantive revision to serve as the authoritative parent document for the new Energy Sector project. Incorporated the complete Energy Sector taxonomy from Design Basis Energy Sector Mindmap v2 (June 2026). Added full GenCo taxonomy; TransCo infrastructure detail; DisCo control systems detail; CntlCo completeness. Added Oil and Natural Gas sub-sector taxonomy. Formalized gas-electric bidirectional cascade as a first-class cross-sub-sector postulated event category. Added sub-sector series inventory table. Added dual-DB architecture acknowledgment. Added Bright Line and Command Broker as cross-sector series requirements. Added series-wide naming convention and references section.
0.3	June 2026	T. Roxey	Section 3.4 (Cross-Sector Consequence Elevation Rule) added. Establishes the governing coordination protocol for DBA-ES-GC-FC4 and DBA-GAS-FC1 at dual-scope compressor station facilities: consequence-perspective separation rule; acceptance criteria assignment rule; and cross-sector consequence elevation rule. Names the NERC CIP gap. Adopts the single-path dependency criterion from FC4 §1.2. Section 5.3 (Nuclear Sector Grid Dependency) expanded; NUC-001-4 added. OI-1 closed.
0.4	June 2026	T. Roxey	Section 5.1 (Water Sector Dependency) expanded to address the bidirectional water-energy interdependency. Section 5.4 (DBA-EN-SF / DBA-W Boundary Protocol) added. Establishes

			instrument assignment rule, forward-direction cross-sector consequence elevation obligation, reverse-direction elevation obligation, and inheritance obligations. OI-5 closed.
0.5	June 2026	T. Roxey	Section 3.4 (Cross-Sector Consequence Elevation Rule) revised to implement the bidirectional rule fully. Named-series elevation trigger table added: DBA-W (bidirectional), DBA-DC (forward only), DBA-MA nuclear (bidirectional). The notification protocol paragraph has been established. Open items OI-DC-1 and OI-MA-1 registered. Gap 1.3 closed at the DBA-EN-SF level.
0.6	June 2026	T. Roxey	Section 4.4 (Consequence Metrics) revised: opening sentence updated from “Three” to “Four” consequence-duration metrics. PPSD (Pipeline Pressure Supply Duration) was added as the fourth series-standard metric. Definition per DBA-GAS-FC2 §1.3; PPSD → GSD sequential relationship established by reference to DBA-EN-UC-1 §5.1. Gap 1.6 closed.
0.7	June 2026	T. Roxey	Section 4.4 (Consequence Metrics) revised: opening sentence updated from “Four” to “Six” consequence-duration metrics. SDD (Supply Delivery Duration) added as the fifth series-standard metric: governing metric for DBA-OIL-FC4 crude and product pipeline transport facility class; established in DBA-OIL-FC4 v0.3 §§1.3 and 4.2; non-additivity rule stated (governing SDD is the longest single-segment critical path); Zone 1/2/3 acceptance criterion framework with T ₁ /T ₂ calibration noted. SABD (Situational Awareness Blackout Duration) added as the sixth series-standard metric: governing metric for DBA-ES-CntlCo functional class; established in DBA-ES-CntlCo v0.2 §1.2; BCC transition non-termination rule stated; Zone 1/2/3 escalation structure with T ₁ /T ₂ calibration noted. Both metrics follow the same sanctioning pathway as PPSD (established at FC level, sanctioned upward to DBA-EN-SF).
0.8	June 2026	T. Roxey	Housekeeping revision. §1.4 Sub-Sector Series Inventory table refreshed across two passes. Initial pass (v0.8 baseline): GC-FC4 corrected to v0.4, DI-FC5 corrected to v0.3, DBA-DC corrected to v1.7, DBA-ES-MA-SF v0.2 added to DBA-ES row; OIL-FC4 corrected to v0.3; DBA-EN Cross-Sector row added (UC-1, UC-2, MA-SF, DBA-VC v0.3). Second pass (N-15/N-16 batch pass closure): all nine DBA-ES FC instrument versions updated to post-batch-pass current versions (HY-FC1 v0.5, TX-FC2 v0.3, TX-FC3 v0.3, GC-FC4 v0.5, DI-FC5 v0.4, CntlCo v0.3, GN-FC6 v0.2, GN-FC7 v0.2, GN-FC8 v0.2); DBA-ES-MA-QB (v0.1) added to DBA-ES row; DBA-ES-MA-

			SF removed from DBA-EN cross-sector row (ES-scoped instrument, correctly listed in DBA-ES row only). Body text references updated to match throughout. §10 series narrative updated to reflect current instrument count and versions. Closes Gap N-1 in DBA-EN Gaps and Recommendations v0.4.
0.9	June 2026	T. Roxey	Parent-registration pass for the marine/strategic terminal promotions and the natural gas and petroleum MA-SF companion layer. §1.4 inventory: DBA-GAS row updated to v1.7 with DBA-GAS-FC-LNG and DBA-GAS-MA-SF (v0.2) added; DBA-OIL row updated to v1.8 with DBA-OIL-FC5 and DBA-OIL-MA-SF (v0.1) added; DBA-ES Sector Framework version corrected v0.8 → v0.9. §4.5 naming convention: promoted-class naming clause added (next sequential number when carved from an existing class; use-case mnemonic when promoted from a named use case); DBA-GAS-FC-LNG and DBA-OIL-FC5 registered. §6.1 Development status refreshed for the gas and petroleum series. Registers the promotions asserted in DBA-GAS-MA-SF v0.2 / DBA-OIL-MA-SF v0.1 and the Sector Framework registrations in DBA-GAS-SF v1.7 / DBA-OIL-SF v1.8.
0.10	June 2026	T. Roxey	Critical Loads subsector (DBA-ES-CL) registered. §1.4 inventory: DBA-DC grouped under the Critical Loads subsector (DBA-ES-CL) within the DBA-ES row; DBA-ES Sector Framework updated v0.9 → v0.10 and DBA-DC updated v1.7 → v1.9 to reflect the instruments issued in this pass. §4.5: note added that a sub-sector may organize facility classes into named subsector groupings (DBA-ES-CL), which do not alter instrument designations. §5.2: DBA-DC identified as founding member of DBA-ES-CL. §6.1 Development status refreshed (DBA-ES v0.10, DBA-DC v1.9). The Critical Loads subsector is defined in DBA-ES §8.5.